PROJECT REPORT TEMPLATE

Dataset Selection, Data Cleaning, Preprocessing, and Model Selection

Student Details

Student Name: _____

Register Number: _____

Course / Class: _____

Project Title: _____

Faculty Guide: _____

Submission Date: _____

1. Project Overview

1.1 Problem Statement

Describe the problem that the project intends to solve.

1.2 Project Objectives

List the major objectives of the project.

1. _____
2. _____
3. _____

1.3 Proposed Approach

Briefly explain how machine learning, deep learning, or data analytics will be used to address the problem.

2. Dataset Description

2.1 Dataset Information

Parameter	Description
Dataset name	
Dataset source / URL	
Domain / application	
Total records / images / samples	
Number of features / attributes	
Target variable / class labels	
Data format	
Dataset licence	

2.2 Dataset Selection Justification

Explain why the dataset is appropriate for the selected problem, considering relevance, availability, size, quality, and suitability.

2.3 Dataset Characteristics

Describe the data types, class distribution, feature characteristics, and known limitations.

3. Data Cleaning

3.1 Data Quality Assessment

Identify potential or observed issues such as missing values, duplicate records, inconsistent formats, invalid entries, outliers, or incorrect labels.

3.2 Data Cleaning Plan

Data Quality Issue	Proposed / Applied Technique	Reason
Missing values		
Duplicate records		
Inconsistent values		
Outliers / invalid entries		
Incorrect labels		

If an issue is not present, state “Not applicable” and provide supporting evidence where possible.

3.3 Cleaning Evidence

Include relevant code snippets, screenshots, or before-and-after summaries for the cleaning work completed so far. Do not include model performance results.

4. Data Preprocessing

4.1 Preprocessing Workflow

Describe the planned sequence of preprocessing operations.

4.2 Preprocessing Techniques

Technique	Purpose	Configuration / Method
Missing-value handling		
Categorical encoding		
Feature scaling / normalization		
Feature selection / extraction		
Image resizing / enhancement (if applicable)		
Text cleaning / tokenization (if applicable)		
Other relevant techniques		

Include only the techniques relevant to your project.

4.3 Preprocessing Justification

Explain why each selected preprocessing technique is appropriate for the dataset and problem.

4.4 Preprocessing Evidence

Include relevant code snippets, workflow diagrams, sample transformations, or screenshots.

5. Preliminary Exploratory Data Analysis (EDA)

Include suitable preliminary analysis, such as:

• Dataset dimensions and summary statistics.

• Data types and feature distributions.

• Class distribution or target-variable distribution.

• Missing-value and duplicate-record summaries.

• Suitable graphs and visualisations.

Key Observations

1. _____

2. _____

3. _____

6. Dataset Splitting Plan

Dataset Subset	Proposed Percentage	Planned Purpose
Training	70%	Model training
Validation	15%	Model and hyperparameter selection
Testing	15%	Final evaluation in the next phase

Explain the planned splitting method, such as random, stratified, or group-based splitting. Describe how data leakage will be prevented.

The percentages may be modified according to the dataset, project requirements, and faculty guidance.

7. Candidate Model Selection

Identify at least three suitable candidate models wherever feasible.

Candidate Model	Model Type	Suitability for the Project
Model 1		
Model 2		
Model 3		

7.1 Model Selection Justification

Explain the theoretical suitability of each candidate model based on:

• Nature of the problem and data.

• Dataset size and characteristics.

• Model complexity and interpretability.

• Expected computational and memory requirements.

• Relevant research literature or established methods.

7.2 Proposed Model

State the model or models proposed for the next phase and explain why they are suitable candidates.

This section records the proposed model selection. Experimental comparison and final selection based on measured performance will be undertaken next week.

8. Proposed Project Workflow

Insert a flowchart illustrating the planned pipeline:

Problem Definition → **Dataset Acquisition** → **Data Quality Assessment** → **Data Cleaning** → **Data Preprocessing** → **Preliminary EDA** → **Dataset Splitting Plan** → **Candidate Model Selection**

9. Conclusion

Summarise the project problem, dataset selected, cleaning and preprocessing approach, and proposed model selection.

Do not include experimental model results or performance-based conclusions.

10. References

List the dataset source, relevant research papers, documentation, and software libraries. Use a consistent referencing style, preferably APA 7th edition.

1. _____

2. _____

3. _____

11. Appendix

Include relevant preliminary code snippets, dataset samples, screenshots, and supplementary documentation.

Student Declaration

I confirm that this report accurately describes the dataset selected and the cleaning, preprocessing, and model-selection work completed or proposed for this stage of the project.

Student Signature: _____

Date: _____